Lintora Case Study – Product Analytics

Product Analytics Data Scientist: Take-Home Exercise

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Investigation Lines & Priority

Based on Lintora's team inputs, I have prioritized the following tracks:

1. **Paid Social Investment (Head of Growth)**: high priority.

High revenue impact, the intention of doubling the budget on the specific channel risks a waste of $100k/month.

2. **Trial Conversion Drop Post-v2 Launch (Head of Product)**: low priority.

Quality metrics proved v2 onboarding is not the root cause of the drop.

3. **WAU Drop in Mid-March (VP Engineer)**: low priority.

The analysis proved the drop was caused by a 88% marketing budget cut.

4. **AI Fix Default Activation (AI Fix PM)**: low priority.

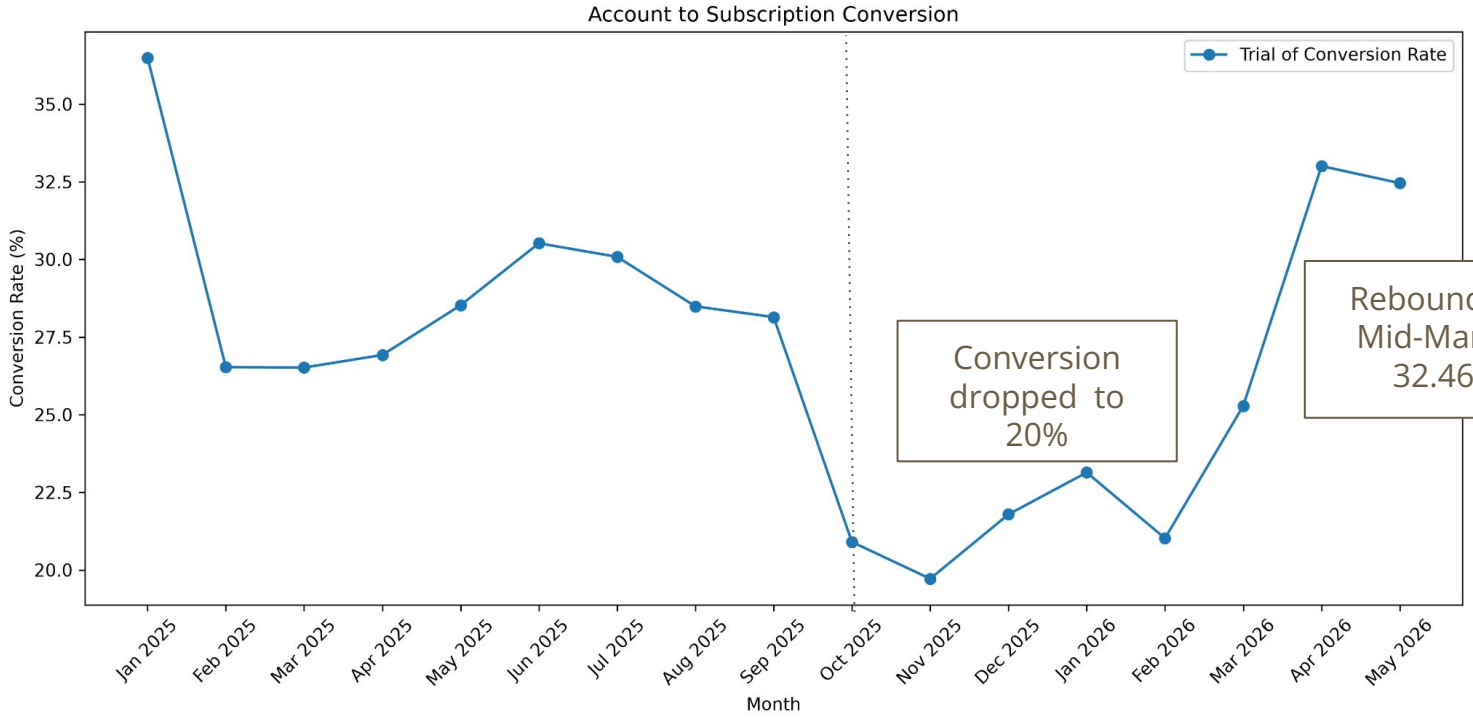
The Machine Learning shows it does not have an impact on the trial conversion.

BONUS: Developed a Random Forest to identify the most impactful user behaviors driving trial conversion.

How is the trial conversion after Oct 1st 2025?

The Head of Product states a decreasing of trial to subscription rate after October 2025, when the version 2 (v2) of the software has been released.

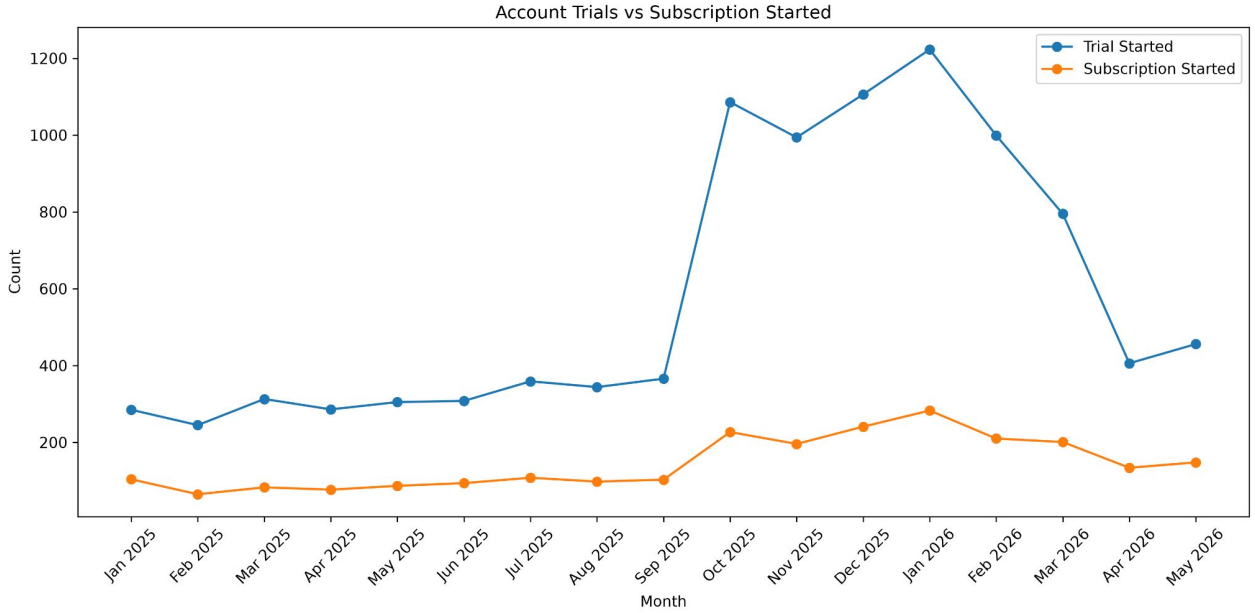
Let's do a **cohort analysis** of the **trial conversion rate** (#account_sub/#account_trial) for each account subscribed.



Is the v2 the root cause ?

Separating the **Trial** from the **Subscribed** population reveals a different story:

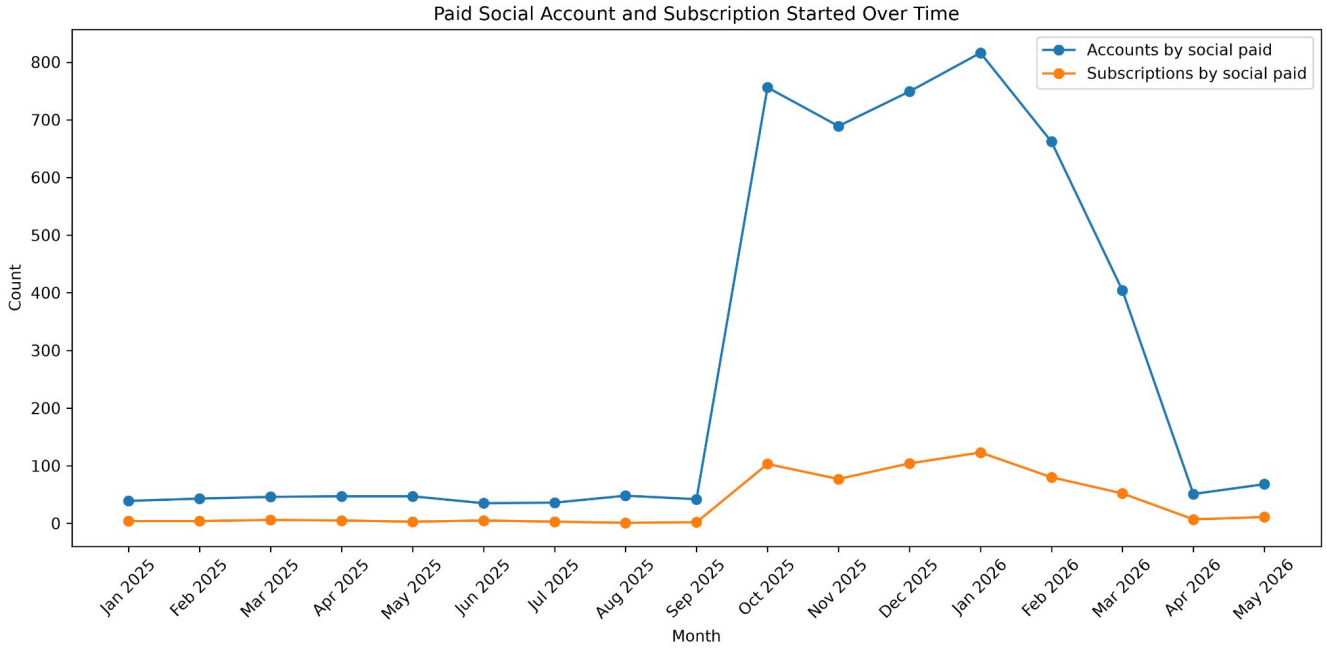
- After September, the number of **trial accounts increased** dramatically, while the **subscribed** user trend remains almost **unaffected**.
- What caused this trend? Is it lower quality accounts from Paid Social, or actual issues within the v2 onboarding?



Double the Paid Social budget is really a good idea?

From October 2025, **Paid Social investments caused an increment of new trial accounts** up to Feb/April 2026, lasting until early 2026.

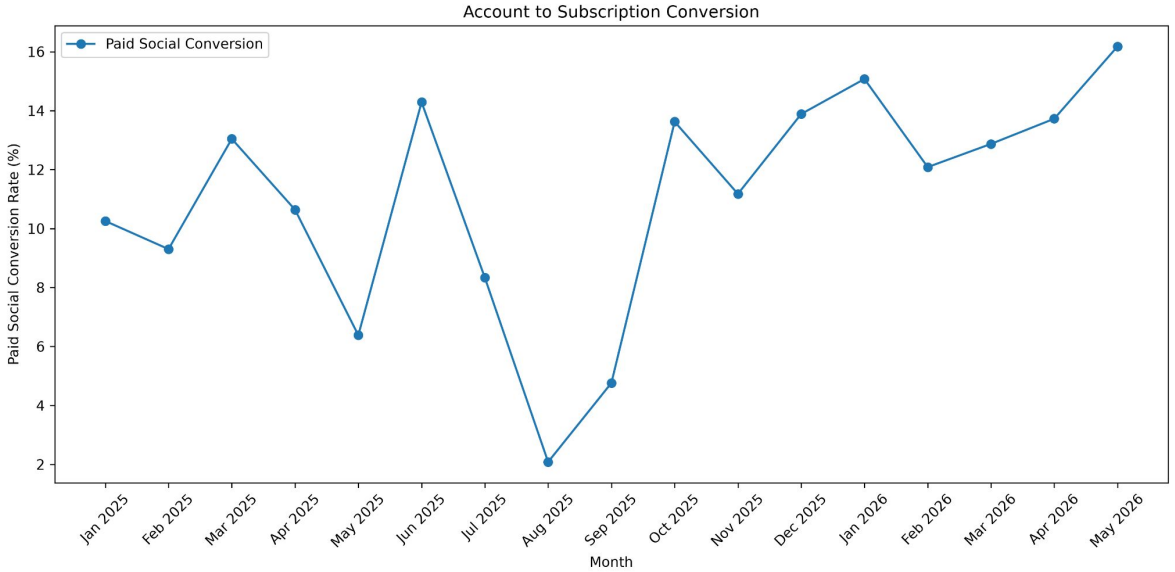
Which was the impact? While trials saw an encouraging increase of almost of 90% in October (42 to 756), the paid subscriptions barely increased.



Double the Paid Social budget is really a good idea?

The **Paid Social conversion rate** has fluctuated between **10** and **15%**, before and after October 2025.

This confirms that the drop in the total conversion rate was caused by the new trial accounts and not by the v2 onboarding.



Conclusion: Paid social shows a significantly lower conversion respect the other channels.
Doubling this budget would heavily **damage** the company's ROI and lead to a clear loss of efficiency

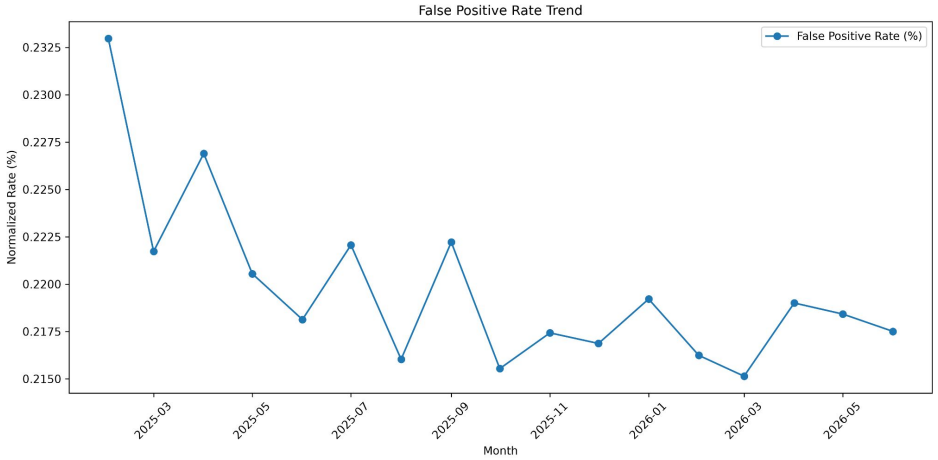
Are there issues with v2?

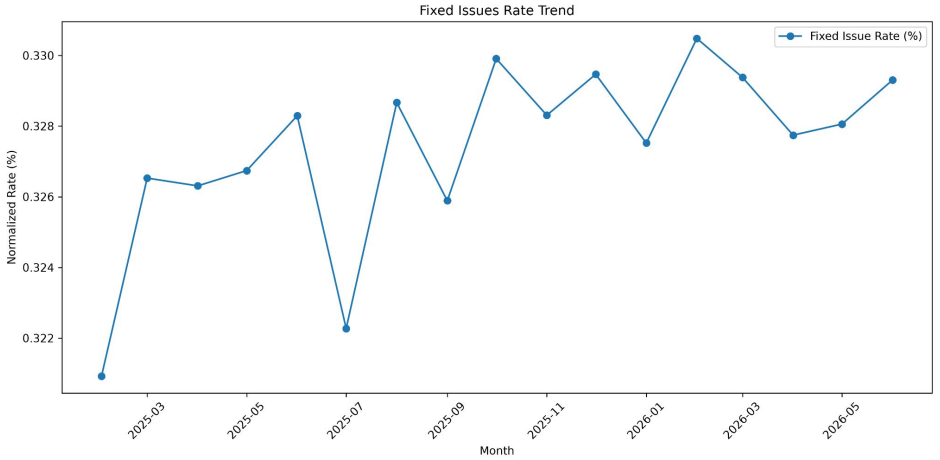
Let’s check the product quality and the performances after v2 onboarding.

Metrics (Normalized on total issues raised):

- **False Positive Rate**: Stable within a healthy 21% – 23% range over time.
- **Fixed Issues Rate**: Consistently solid, fluctuating between 32% – 33%.

Conclusion: **no sign of product degradation or regression after the v2 launch.**
→ The software-v2-issue can be cleared by all charges.

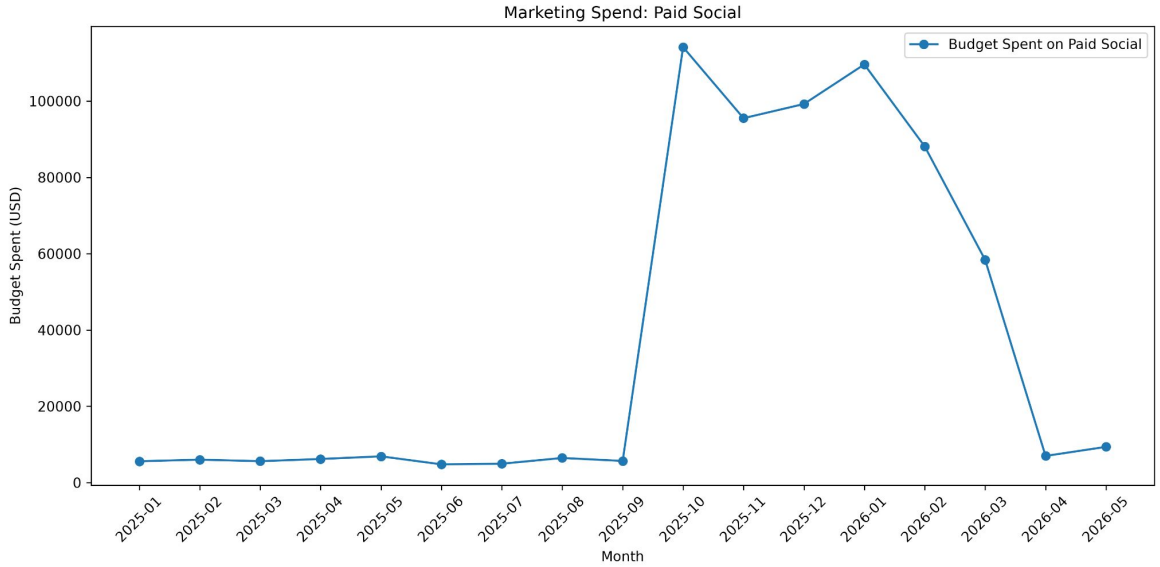




Investigation of the mid-March drop

What the VP Engineer observed is real, but it is caused by **fewer new users coming from marketing**, not a technical software problem.

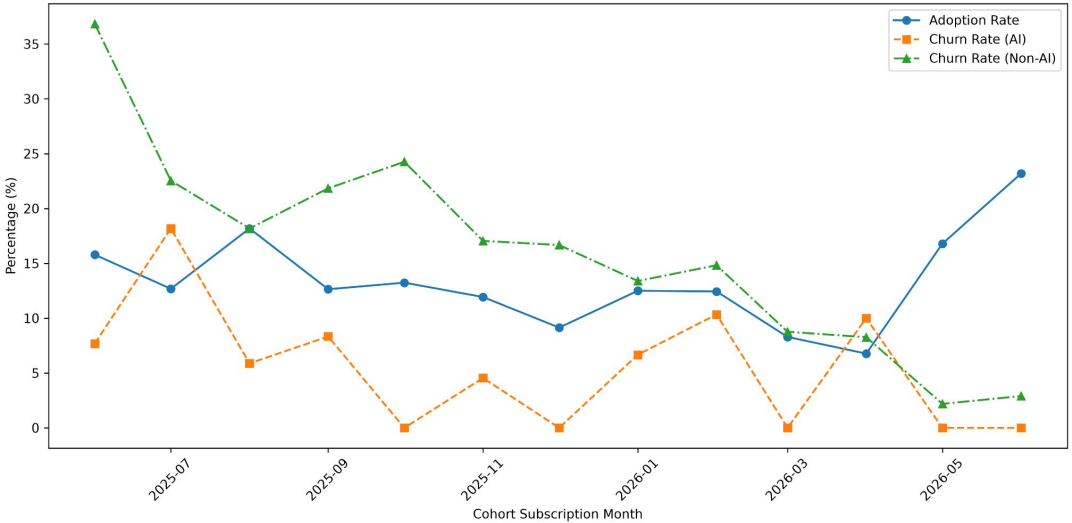
Data from marketing spend confirms an **88% reduction** in the Paid Social budget between March and April 2026.



Room for improvement: **internal communication** between Marketing and Engineering could prevent future misinterpretations by technical leaders.

AI Fix churn

We can look at the cohort analysis after June 2025 (AI fixed was launch) and check the **adoption rate** of ai users, **churn rate** of ai users and no-ai users.



- The **AI churn rate** is lower than the Non-AI one and remains quite stable, indicating the AI Fix users are high-engaged users.
- The **Non-AI churn rate** decreases drastically over time. However, the population is biased due to the influx of low-intent Paid Social accounts in 2025.
- The increase of the adoption rate after March 2026 is an **illusion**: caused by the 88% Paid Social budget cut.

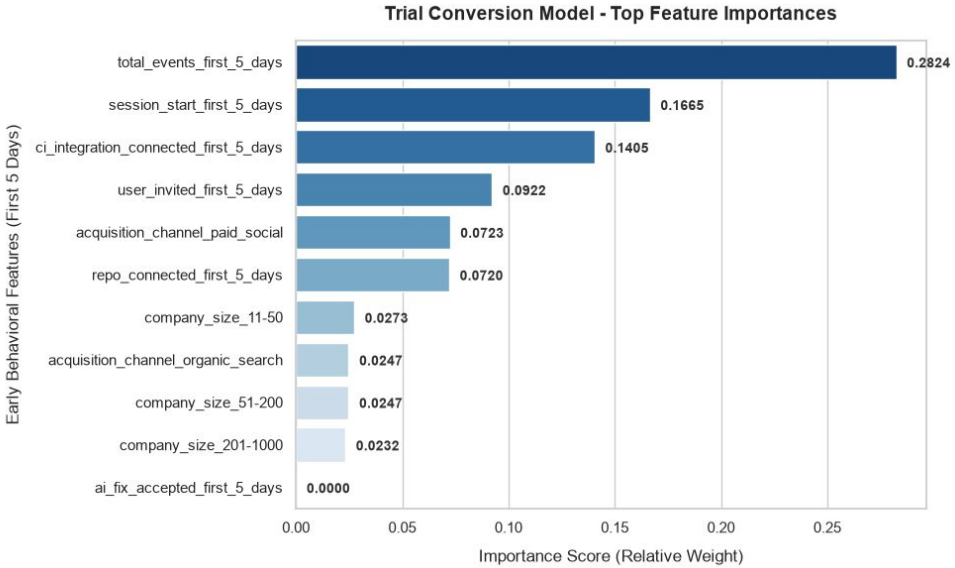
Conclusion: the data suffers of bias and are not conclusive. I recommend running an A/B test on independent cohorts to isolate the true impact of the AI Fix.

Bonus: Machine Learning

What are the most critical signals in the first 5 days that predict whether a **trial account will convert**?

Model Architecture: Random Forest Classifier (100 trees, 80/20 train/test split, stratified on target).

- **Predictive Accuracy**: The classifier achieved a 80.2% accuracy in predicting 14-day trial conversions.
- **Primary Conversion Drivers**: User activity (total_events, session_start at ~44%) and deep technical setup (ci_integration, repo_connected a ~21%) are the most significant features.
- **Marketing Quality Indicator**: acquisition_channel_paid_social holds a strong ~7% importance score.
- **The AI Verdict**: ai_fix_accepted holds 0.00% importance in the model, proving that early AI interactions do not influence conversion outcomes.
- **Next Steps**: Improve the model's ability to detect paid subscribers by balancing the dataset.



Recommandations & Next Steps

- **Marketing Optimization (High Priority):** Restructure the Paid Social campaigns and stop inefficient spending.
    - Reallocating this budget to other channels (which have a 30% conversion rate vs. Paid Social's 12%) will save up to $100,000/month in wasted spend, delivering a 2.5x increase in conversion efficiency.
- Improve the **team communication**: the mid-March WAU drop was an intentional marketing budget cut, not an infrastructure issue. Need to establish some dashboard of communication inside the team.
- **AI Fix Default Activation (Low priority):** reject the proposal of the default AI fix activation. The Machine Learning proved that the impact on the conversion in 0%.
    - **Action Plan:** Design a randomized A/B Test on independent user cohorts to safely isolate the true value of the AI Fix feature without selection bias.